

The Evolution of Irish Medium Educational Resources: A Comprehensive Analysis of the University of Galway's MGO Programme and the Digital Pedagogies of Seán Ó Grádaigh

Executive Summary

The intersection of Irish-medium education, initial teacher education (ITE), and mobile technology represents a dynamic and critical niche within the broader landscape of Irish education. This report provides an exhaustive analysis of the digital resources, pedagogical strategies, and institutional frameworks associated with the **Máistir Gairmiúil san Oideachas (MGO)** programme at the **University of Galway** (formerly National University of Ireland, Galway). Specifically, it investigates the portfolio of work developed under the leadership of **Seán Ó Grádaigh**, a pivotal figure in this domain, and addresses the user's inquiry regarding "lost" resources, particularly chemistry experiments and interactive iBooks that were once prominent but have since become difficult to locate.

The research indicates that the resources in question—specifically the **"Sci-Hook"** series and the **"Ceimic"** (Chemistry) iBooks—were the product of a specific era of technological innovation (2012–2016) heavily tied to the Apple ecosystem. These resources utilized the "Hook" pedagogical model to engage students in science topics. While they remain available on specific platforms like Apple Books, their visibility on the open web has diminished due to the deprecation of platforms like iTunes U and the shifting focus towards open educational resources (OER). Furthermore, a parallel resource stream known as **Kitchen Chemistry** (or *Ceimic Chistine*) has been identified, which targets similar scientific concepts through a different pedagogical lens.

This report traces the trajectory of these resources from their inception as "Design-Based Research" projects to their current status. It analyzes the shift from product-creation (static iBooks) to process-oriented training (student teachers as content creators) and evaluates the impact of funding bodies such as **An Chomhairle um Oideachas Gaeltachta agus Gaelscolaíochta (COGG)** and the **Apple Distinguished School** designation on the sustainability and availability of these materials.

1. Institutional Ecosystem: The University of Galway and Irish Medium Education

To understand the nature, scope, and availability of the resources in question, it is essential to first understand the institutional ecosystem in which they were created. The University of Galway holds a unique statutory and cultural position regarding the Irish language, which fundamentally shapes its output in educational technology.

1.1 The Strategic Mandate of the School of Education

The School of Education at the University of Galway is not merely a provider of teacher training; it is the primary hub for **Irish Medium Education (IME)** teacher preparation in Ireland. This role is driven by the university's proximity to the Connemara Gaeltacht and its legislative mandate to support the language.¹

The strategic objective of the School is to address the chronic supply and demand issues facing *Gaelcholáistí* (Irish-medium secondary schools) and Gaeltacht schools. These schools often struggle to find qualified teachers who are proficient in both their subject matter (e.g., Chemistry, Physics, Mathematics) and the target language (Gaeilge). Consequently, the resources produced by the School are designed to serve a dual purpose: training the teacher in subject-specific pedagogy while simultaneously providing high-quality, Irish-language instructional materials to a sector that is historically under-resourced.³

1.2 The Máistir Gairmiúil san Oideachas (MGO) Programme

The **Máistir Gairmiúil san Oideachas (MGO)** is the flagship programme driving this resource creation. Established in 1998, it is the only two-year, full-time Professional Master in Education (PME) offered entirely through the medium of Irish.¹

1.2.1 Programme Structure and Philosophy

The MGO is designed as an immersion programme. It prepares teachers for the second-level sector, with a specific focus on immersion pedagogies. The programme's philosophy is rooted in the belief that effective IME requires more than just translating English resources; it requires the creation of bespoke materials that reflect the linguistic and cultural reality of the Irish classroom.

The programme is recognized by the **Teaching Council of Ireland** for registration purposes, ensuring that its graduates are fully qualified for the post-primary sector.¹ It boasts the highest employment rate of any teacher education programme in Ireland, a statistic that underscores the high demand for graduates who possess the unique blend of technological competence and linguistic fluency cultivated by the course.¹

1.2.2 The "Student-as-Producer" Model

A critical insight revealed by the research is that the resources the user is looking for—the chemistry experiments and iBooks—were not solely produced by academic staff. Instead, the

MGO operates on a "Student-as-Producer" model. Cohorts of student teachers, under the supervision of lecturers like Seán Ó Grádaigh and Dr. Veronica McCauley, develop these resources as part of their coursework.⁶

This pedagogical approach serves two functions:

1. **Skill Acquisition:** It teaches student teachers how to use educational technology (EdTech) tools like **iBooks Author**, **iMovie**, and **Keynote** to create professional-grade learning materials.
2. **Resource Generation:** It creates a repository of materials (like the *Ceimic* books) that can be disseminated to the wider teaching community, filling the resource gap in the IME sector.⁶

1.3 Strategic Partnerships and Funding

The sustainability of these initiatives relies on key partnerships. The research highlights the role of **COGG** (An Chomhairle um Oideachas Gaeltachta agus Gaelscolaíochta), which funds bursaries for MGO students (€2,000 per year) and supports the development of teaching materials.⁴ Additionally, the university's status as an **Apple Distinguished School** (discussed in Section 2.3) has facilitated access to hardware and software that defined the format of the resources produced between 2012 and 2016.

2. Seán Ó Grádaigh: The Architect of Mobile Pedagogy

The user's query explicitly names **Seán Ó Grádaigh**. Understanding his academic profile, research interests, and technological philosophy is key to locating the resources and understanding why they exist in their specific formats.

2.1 Academic Profile and Research Interests

Seán Ó Grádaigh is a Lecturer in Education at the University of Galway and the current Programme Director of the MGO (since 2017).³ His background combines second-level teaching (Gaeilge, Geography, and History) with a specialization in educational technologies. His research fingerprint³ identifies several core clusters of interest:

- **Mobile Learning (m-Learning):** Investigating how mobile devices can enhance teaching and learning.
- **Teacher Education:** Specifically, the training of pre-service teachers in digital competencies.
- **Computational Thinking:** Recent work (2021) explores using mobile learning to support computational thinking in ITE.⁸
- **Irish Medium Education:** The context in which all his technological interventions are applied.

2.2 Pedagogical Philosophy: "Content Creation"

Ó Grádaigh's philosophy centers on the idea that teachers should be creators, not just

consumers, of digital content. In interviews and publications, he emphasizes that the depth of learning is enhanced when students (both student-teachers and school pupils) design their own materials.⁹

For example, in the creation of the *Ceimic* resources, Ó Grádaigh noted that "Geography students rarely get the opportunity to experience the great outdoors, but this type of technology brings the living Geography into the classroom," and that for the science resources, the goal was for student teachers to "understand how to use technology effectively... to create learning materials that engage all types of learners".⁶

2.3 The Apple Distinguished Educator (ADE) Connection

A pivotal factor in the format of the "lost" resources is Seán Ó Grádaigh's status as an **Apple Distinguished Educator (ADE)**, a recognition he received in 2013.³ The ADE program recognizes K-12 and higher education pioneers who are using Apple technology to transform teaching and learning.

Consequently, the MGO programme became heavily integrated with the Apple ecosystem:

- **1:1 iPad Deployment:** All MGO students and staff are equipped with iPads.⁵
- **Apple Distinguished School:** The MGO programme was the first educational programme outside the US to be recognized as an Apple Distinguished Programme (now School) in 2015, a status renewed for 2024–2027.¹
- **Format Implications:** Because of this affiliation, the primary output format for resources developed during the 2012–2016 peak was the **iBook** (using the .ibooks or .iba file format). These files are proprietary to Apple and can typically only be viewed on iPads or Macs via the Apple Books app. This explains why users looking for standard PDFs or web pages might struggle to find them today—they are locked inside a specific technological ecosystem.⁵

3. The "Lost" Resources: The Sci-Hook Series and "Ceimic" iBooks

The core of the user's request concerns "leaving certificate resources like chemistry experiments" that were once found but are now missing. Based on the research snippets, these resources can be positively identified as the **Sci-Hook Series** (Science Hooks) and their associated **iBooks**, specifically the volume titled **Ceimic**.

3.1 The Sci-Hook Series: Concept and Pedagogy

The **Sci-Hook Series** was a Science Education Research (SER) project launched around 2012, led by Dr. Veronica McCauley and Seán Ó Grádaigh.¹¹

3.1.1 The Pedagogical "Hook"

The central concept of this series is the "Hook." In educational theory, a hook is an instructional technique used at the beginning of a lesson to stimulate student attention,

interest, and engagement.¹² The researchers identified that science students often disengage because they see the subject as abstract or disconnected from reality.

To combat this, the team designed "Sci-Hooks"—short, introductory video clips (typically under 2 minutes) that capture the "engaging and novel aspects" of a science topic.⁶ The videos utilize:

- **Quirky Film Editing:** To maintain visual interest.
- **Animation Fluidity:** To explain abstract concepts.
- **Curious Content:** Focusing on anomalies, challenges, or "messy" science to provoke questions.¹¹

3.1.2 Target Audience: Junior vs. Leaving Certificate

The user mentions "leaving certificate resources," but the research indicates that the Sci-Hooks were primarily designed for the **Junior Science** curriculum (ages 12–15).¹¹ However, the snippets also clarify that because the concepts are fundamental (e.g., atomic structure, chemical reactions), they "can be extended to Senior Cycle also".¹⁴ This likely explains the user's memory of using them for Leaving Certificate chemistry; the content is relevant even if the original target was younger students.

3.2 The "Ceimic" (Chemistry) iBooks

The videos were not just released as standalone files; they were embedded into interactive **Multi-Touch iBooks**. This is likely the "multiple iBooks" the user refers to.

3.2.1 "Ceimic" (The Irish Resource)

The specific resource for Irish-medium chemistry is titled **Ceimic**.

- **Authors:** Seán Ó Grádaigh & Veronica McCauley.¹⁵
- **Release Date:** July 30, 2012.¹⁶
- **Format:** Apple iBook (Multi-Touch).
- **Content:** The book contains over 10 embedded videos. Each chapter includes:
 - **The Hook Video:** An embedded video file.
 - **Explanatory Text:** Providing the scientific theory behind the hook.
 - **Methodology:** Instructions on how to carry out the investigation.
 - **Pedagogical Guidance:** Advice for teachers on how to incorporate the hook into a lesson.
 - **Assessment:** Associated questions and cross-curricular links (literacy & numeracy).¹⁶

3.2.2 The English Equivalents

Parallel to *Ceimic*, English-language versions were released:

- **Chemistry Resource Hooks.**¹⁶
- **Physics Resource Hooks.**¹⁷
- **Biology.**¹⁷

3.2.3 "Fisic" (The Physics Resource)

There is also a confirmed Irish-language resource for Physics titled **Fisic**, authored by the same team.⁶

3.3 Availability: Where Did They Go?

The user states, "I can't find them anymore." The research suggests several reasons for this disappearance from the "surface web":

1. **Platform Specificity:** As iBooks, these resources are not indexed by standard search engines in the same way PDFs or HTML pages are. They live within the **Apple Books Store**. To find them, one must search within the Apple Books app, not Google.
2. **Deprecation of iTunes U:** The University of Galway previously hosted these videos on iTunes U. Apple discontinued iTunes U in late 2021. Any direct links to iTunes U collections would now be dead.⁶
3. **Website Link Rot:** The project originally had a dedicated website at www.sciencehooks.scoilnet.ie.¹⁴ While Scoilnet is active, specific subdomains or microsites often go offline or are archived when funding concludes. The snippets indicate the site was a "satellite website" maintained under the Scoilnet initiative, but its current active status is ambiguous.¹⁸

3.4 Actionable Retrieval Strategy

Despite these challenges, the resources **are** still available if the correct channels are used:

- **Apple Books:** The *Ceimic* and *Chemistry Resource Hooks* books are listed as available for free download on the Apple Books store. The user needs an Apple ID and an iOS/macOS device to access them.¹⁵
- **Scoilnet Archives:** While the microsite might be tricky, the individual video assets are often cataloged in the main Scoilnet database. Searching for "Science Hooks" or "Seán Ó Grádaigh" on scoilnet.ie is a recommended strategy.¹⁹

4. The "Kitchen Chemistry" / "Ceimic Chistine" Parallel

The user's query about "chemistry experiments" might also refer to a separate but related project called **Kitchen Chemistry** (or *Ceimic Chistine* in Irish). It is crucial to distinguish this from the *Sci-Hooks* project, as they are often conflated.

4.1 Origin and Collaboration

Kitchen Chemistry began in 2010 as an outreach venture by the University of Galway's **School of Chemistry** (not Education). It was led by Dr. Nicole Walshe and a team of postgraduate chemistry students.²⁰

- **Evolution:** In 2012, Dr. Walshe collaborated with Dr. Veronica McCauley (School of Education) to formalize these outreach experiments into a digital resource. This

collaboration bridged the gap between pure science (Chemistry Department) and pedagogy (Education Department).²⁰

4.2 Content and Scope

- **Philosophy:** The project promotes science using materials found in a typical kitchen (e.g., vinegar, baking soda, red cabbage indicator) to make chemistry accessible and non-threatening.
- **Format:** A series of six videos and a "Teacher Resource Pack".²¹
- **Language:** The collection is available *trí Ghaeilge* as **Ceimic Chistine**, making it another vital resource for the user.²¹
- **Target:** Primarily aimed at primary school students (to "stimulate curiosity and imagination"), but the principles are fundamental chemical concepts relevant to Junior Cycle.²⁰

4.3 Key Differences: Sci-Hooks vs. Kitchen Chemistry

Feature	Sci-Hook Series (Ceimic)	Kitchen Chemistry (Ceimic Chistine)
Lead Department	School of Education (Seán Ó Grádaigh)	School of Chemistry & Education
Target Audience	Junior Science (12-15 yrs) / Senior	Primary / Early Secondary
Pedagogy	The "Hook" (Attention Grabbing)	Hands-on, Home-based experimentation
Format	iBooks (Multi-Touch), Embedded Video	Web Videos, PDF Teacher Packs
Key Authors	Ó Grádaigh, McCauley	Walshe, McCauley, Conway
Availability	Apple Books (iOS/Mac)	Scoilnet, YouTube, University Website

5. How Resources Have Changed Over Time (2012–2024)

The user asks specifically "how they changed over time." The trajectory of these resources mirrors the broader evolution of educational technology, shifting from "Device-Centric" creation to "Cloud-Centric" and "Open Access" models.

Phase 1: The "App & iBook" Era (2012–2016)

- **The Trend:** This period was defined by the novelty of the iPad. The focus was on creating "apps" or interactive "books" that lived on the device.
- **The Output:** *Ceimic*, *Fisic*, and the *Sci-Hook* iBooks.

- **Characteristics:** These resources were "walled garden" products. They were highly polished, interactive, and media-rich, but required specific hardware (Apple) to function. The "content creation" ethos meant students were building heavy, complex files.¹⁰
- **User Experience:** For the user, this was the "golden age" of finding distinct, downloadable "products" like the chemistry experiments iBook.

Phase 2: The "Coding & Computational Thinking" Era (2017–2019)

- **The Trend:** As the novelty of digital books wore off, the focus shifted to *what students could do* with the devices beyond reading.
- **The Shift:** Seán Ó Grádaigh's research during this time pivoted towards **Mobile Learning for Computational Thinking**.⁸
- **The Output:** Resources likely shifted away from subject-specific content (like chemistry hooks) towards process-based resources: coding tutorials, Swift Playgrounds, and logical problem-solving activities. The "resource" became the *activity* of coding, rather than a book about chemistry.

Phase 3: The "Remote & Hybrid" Era (2020–Present)

- **The Trend:** The COVID-19 pandemic forced a radical rethink of digital resources. The priority shifted from "engagement" (hooks) to "continuity of learning" and "remote observation."
- **The Shift:** Recent research by Ó Grádaigh focuses on **Virtual Professional Experience Observation (VPEO)** and synchronous hybrid approaches.²²
- **The Output:** The "latest resources" are no longer just for students; they are for *teachers*. They include protocols for recording classes, using video for reflection, and hybrid classroom management.²³
- **Platform Agnosticism:** There is a move away from proprietary files towards universal platforms. Snippets mention students "sharing through Google Drive".²³ This makes resources more accessible but less "packaged." A teacher today is more likely to share a folder of video clips via Google Drive than to author a compiled iBook.

Phase 4: The "Open Educational Resource" (OER) Era (Current)

- **The Trend:** There is a growing institutional push for **Open Educational Resources (OER)** that are free, editable, and platform-neutral.
- **The Output:** The University of Galway Library has launched an **Open Press** initiative.²⁴ This platform allows staff and students to publish textbooks that are web-accessible (e.g., using Pressbooks), solving the "locked in an iBook" problem. While *Ceimic* was a precursor, the future lies in these open platforms.

6. The Latest Resources and Current Landscape

For the user seeking the "latest resources," the landscape has diversified. It is no longer just about a single author or a single file type.

6.1 Current MGO Student Outputs

The "Student-as-Producer" model continues. MGO students are currently assessed on creating:

- **Education Posters**
- **Videos**
- **Podcasts**
- **Digital Books** ⁵

These materials are often disseminated locally within schools or shared via professional networks (like Twitter/X or Scoilnet) rather than being published as formal books. This decentralization makes them harder to "find" as a single collection but richer in volume.

6.2 The MGO Bursary Scheme

A significant resource *for* the students (rather than *by* them) is the extended bursary scheme.

- **Detail:** All MGO students now receive **€2,000 per year**, funded by COGG.⁴
- **Implication:** This funding is intended to secure the pipeline of Irish-medium teachers, ensuring that the creation of Irish-language resources continues by supporting the people who make them.

6.3 Digital Observation Resources

The most cutting-edge resources developed by Seán Ó Grádaigh's team relate to **teacher training via video**.

- **EVO (Emergency Virtual Observation):** Protocols for observing student teachers remotely.
- **Video Reflection:** Tools and guides for teachers to record their own practice for self-improvement.³

6.4 Accessing the "Ceimic" Material Today

To summarize for the user, here is the status of the specific items:

Resource	Status	How to Access
Ceimic (iBook)	Active	Download Apple Books app -> Search "Ceimic".
Chemistry Resource Hooks	Active	Download Apple Books app -> Search Title.
Science Hooks Website	Archived/Uncertain	Search Scoilnet.ie for "Science Hooks".
Kitchen Chemistry	Active	Search Scoilnet.ie or University of Galway Chemistry Dept.
Videos (Standalone)	Scattered	Search YouTube (Physics

		Hooks Channel) or Scoilnet.
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7. Strategic Implications and Conclusion

The "lost" resources the user is looking for are not truly lost; they are artifacts of a specific technological moment—the "iBook era" of 2012–2016. Led by **Seán Ó Grádaigh** and **Veronica McCauley**, and driven by the University of Galway's mandate for Irish Medium Education, these resources (the *Sci-Hook* and *Ceimic* series) represented a pioneering attempt to create bespoke, high-quality digital textbooks for the Irish curriculum. Their apparent disappearance is due to the shift in platforms (iTunes U to Apple Books/Podcasts) and the evolution of the web (link rot of satellite sites). However, the pedagogical core—using "hooks" to engage students and empowering student-teachers to create content—remains central to the MGO programme.

The latest resources have shifted form. They are less likely to be monolithic iBooks and more likely to be:

1. **OERs** published via Open Press.
2. **Cloud-shared assets** (Google Drive/OneDrive) used in hybrid teaching.
3. **Professional Development tools** for remote observation and reflection.

For the user, the path forward involves engaging with the **Apple Books** ecosystem to retrieve the classic *Ceimic* texts and monitoring **Scoilnet** and the University's **Open Press** for the next generation of materials. The MGO programme, fortified by renewed Apple Distinguished School status and COGG funding, remains the engine room for this innovation, ensuring that the supply of Irish-medium resources will continue to evolve, even if the formats change.

Table 1: Detailed Inventory of Irish Language Resources (University of Galway)

Title / Project	Authors / Leads	Format	Description	Current Availability
Ceimic	Seán Ó Grádaigh, V. McCauley	iBook (Multi-Touch)	Irish-language chemistry hooks, theory, and methodology.	Apple Books (Free) ¹⁵
Fisic	Seán Ó Grádaigh, V. McCauley	iBook (Multi-Touch)	Irish-language physics hooks.	Apple Books (Free) ⁶
Chemistry Resource Hooks	V. McCauley, S. Ó Grádaigh	iBook (Multi-Touch)	English version of the Chemistry hooks.	Apple Books (Free) ¹⁶
Ceimic Chistine	N. Walshe, V.	Video / PDF Pack	"Kitchen	Scoilnet /

	McCauley		Chemistry" experiments using household items.	University Website ²¹
The Story of 1916 (Scéal 1916)	Seán Ó Grádaigh	Digital Book	Bilingual resource on the 1916 Rising.	Apple Books ²⁵
Digital Reflection on iPad	Seán Ó Grádaigh	Digital Book	Guide on using iPad for reflective practice.	Apple Books ²⁵
Geography Field Investigation	Seán Ó Grádaigh	Digital Book	Resources for Geography fieldwork (Rivers/Coasts).	Apple Books ²⁵
MGO Programme Resources	MGO Students	Various	Student-created posters, videos, and books.	Distributed via schools/Scoilnet

Table 2: Timeline of Resource Evolution

Period	Focus	Key Technology	Key Outputs
2012–2016	Content Creation	iPad, iBooks Author, iTunes U	<i>Ceimic, Fisic, Sci-Hooks, Kitchen Chemistry</i>
2017–2019	Computational Thinking	Swift Playgrounds, Coding Apps	Coding resources, m-Learning studies
2020–2022	Remote Continuity	Zoom, Teams, Video Observation	EVO protocols, Hybrid teaching guides
2023–Present	Open Access (OER)	Open Press, Cloud Platforms	OER Textbooks, Digital Portfolios

Works cited

1. October - University of Galway Masters programme recognised as Apple Distinguished School, accessed January 27, 2026, <https://www.universityofgalway.ie/about-us/news-and-events/news-archive/2024/october/university-of-galway-masters-programme-recognised-as-apple-distinguished-school.html>
2. Education (Professional Master) - University of Galway, accessed January 27, 2026, <https://www.universityofgalway.ie/courses/taught-postgraduate-courses/education-on-professional-master.html>
3. Seán Ó Grádaigh - University of Galway Research, accessed January 27, 2026, <https://research.universityofgalway.ie/en/persons/se%C3%A1n-%C3%B3-gr%C3%A1daigh/>
4. November - University of Galway announce bursary scheme for student

teachers, accessed January 27, 2026,

<https://www.universityofgalway.ie/about-us/news-and-events/news-archive/2022/november/university-of-galway-announce-bursary-scheme-for-student-teachers-.html>

5. Máistir Gairmiúil san Oideachas (MGO) - University of Galway, accessed January 27, 2026,
<https://www.universityofgalway.ie/media/nuigalwayie/content/files/courses/uploads/ADSApplication.pdf>
6. Apple Regional Training Centre - University of Galway, accessed January 27, 2026,
<https://www.universityofgalway.ie/colleges-and-schools/arts-social-sciences-and-celtic-studies/education/educational-technology-centre/apple-training/>
7. Seán Ó GRÁDAIGH | Director of MGO Teacher Education programme - ResearchGate, accessed January 27, 2026,
<https://www.researchgate.net/profile/Sean-O-Gradaigh>
8. (PDF) Mobile Learning to Support Computational Thinking in Initial Teacher Education: A Case Study - ResearchGate, accessed January 27, 2026,
https://www.researchgate.net/publication/349937647_Mobile_Learning_to_Support_Computational_Thinking_in_Initial_Teacher_Education_A_Case_Study
9. NUI Galway Masters Programme awarded Apple Distinguished Programme for second year running - Galway Advertiser, accessed January 27, 2026,
<http://www.advertiser.ie/galway/article/87243/nui-galway-masters-programme-awarded-apple-distinguished-programme-for-second-year-running>
10. Student Teachers Reimagine the Textbook - News - University of Galway, accessed January 27, 2026,
<https://www.universityofgalway.ie/colleges-and-schools/arts-social-sciences-and-celtic-studies/education/news/student-teachers-reimagine-the-textbook.html>
11. November 2012 - University of Galway Creates New Innovative Resources for the Science Classroom, accessed January 27, 2026,
<https://www.universityofgalway.ie/about-us/news-and-events/news-archive/2012/november2012/nui-galway-creates-new-innovative-resources-for-the-science-classroom-1.html>
12. (PDF) Collaborative lesson hook design in science teacher education: advancing professional practice - ResearchGate, accessed January 27, 2026,
https://www.researchgate.net/publication/287359388_Collaborative_lesson_hook_design_in_science_teacher_education_advancing_professional_practice
13. Collaborative lesson hook design in science teacher education - University of Galway Research Repository, accessed January 27, 2026,
<https://researchrepository.universityofgalway.ie/bitstreams/aa0dd42f-28d4-41b8-b4b9-33b81b3b5a26/download>
14. Sci-hooks, accessed January 27, 2026,
<https://www.ista.ie/wp-content/uploads/2016/07/Sci-hooks.pdf>
15. Ceimic by Seán Ó Grádaigh & Veronica McCauley on Apple Books, accessed January 27, 2026, <https://books.apple.com/us/book/ceimic/id566276039>
16. Chemistry Resource Hooks by Veronica McCauley & Seán Ó Grádaigh on Apple Books, accessed January 27, 2026,

- <https://books.apple.com/us/book/chemistry-resource-hooks/id549454988>
17. Physics Resource Hooks by Veronica McCauley & Seán Ó Grádaigh on Apple Books, accessed January 27, 2026,
<https://books.apple.com/us/book/physics-resource-hooks/id549475578>
18. Scoilnet Websites, accessed January 27, 2026,
<https://www.scoilnet.ie/scoilnet-services/scoilnet-websites/>
19. Science - Scoilnet, accessed January 27, 2026,
<https://www.scoilnet.ie/go-to-primary/theme-pages/science/>
20. November 2016 - University of Galway's Kitchen Chemistry Launch Free Online Resource for Primary Science Classrooms, accessed January 27, 2026,
<https://www.universityofgalway.ie/about-us/news-and-events/news-archive/2016/november2016/nui-galways-kitchen-chemistry-launch-free-online-resource-for-primary-science-classrooms.html>
21. Kitchen Chemistry - Scoilnet, accessed January 27, 2026,
<https://www.scoilnet.ie/go-to-primary/theme-pages/science/kitchenchemistry/>
22. Educ. Sci., Volume 14, Issue 9 (September 2024) – 114 articles - MDPI, accessed January 27, 2026, <https://www.mdpi.com/2227-7102/14/9>
23. From iTE to NQT: Evaluating newly qualified teachers use of mobile technology in their first two years of teaching - University of Galway Research Repository, accessed January 27, 2026,
<https://researchrepository.universityofgalway.ie/bitstreams/48f165fc-1022-45aa-a61a-a598e07622b6/download>
24. Open Press at the University of Galway – Open Access Books and Educational Resources Publishing, accessed January 27, 2026,
<https://openpress.universityofgalway.ie/>
25. MGO by Seán Ó Grádaigh on Apple Books, accessed January 27, 2026,
<https://books.apple.com/ie/book/mgo/id1066781528>